



BITS Pilani presentation

BITS Pilani
Pilani Campus

Faculty Name
Faculty Department



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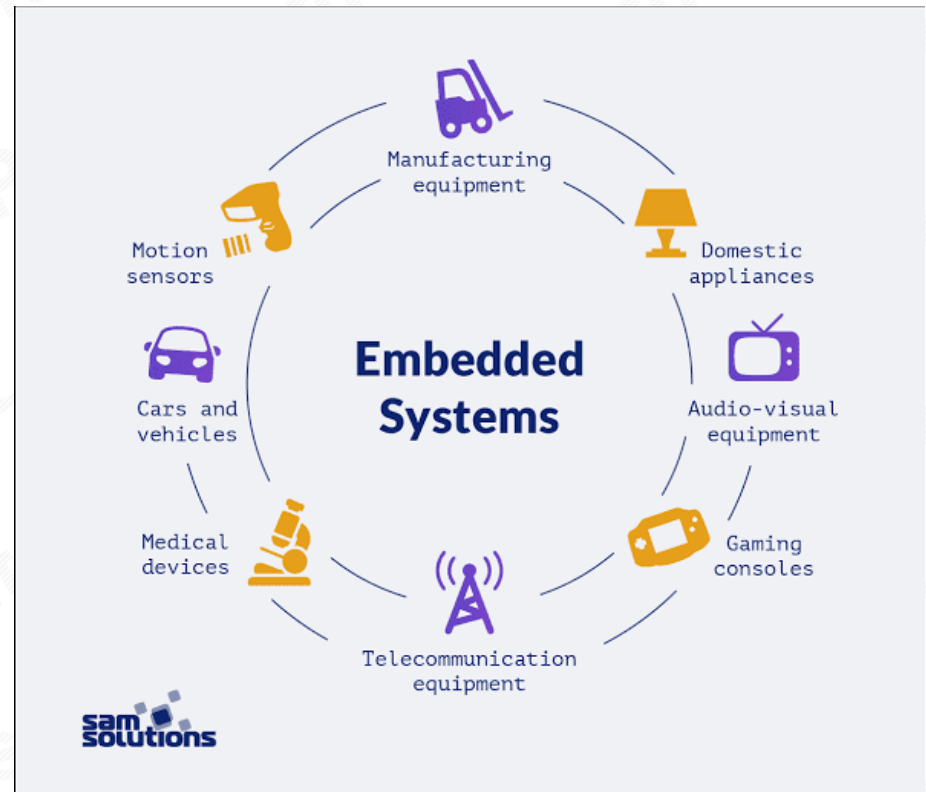
EEE ZG512 / SS ZG512, EMBEDDED SYSTEM DESIGN

Lecture No. 01

Introduction to Embedded System



A system in which software is embedded in to hardware is termed as embedded systems.



Introduction to Embedded System



Designed for a specific task or application.

System is optimized to achieve required specifications which includes.

- ✓ Execution optimization i.e. time required for completing the task allocated.
- ✓ Memory optimization i.e. to reduce the code length.
- ✓ Power optimization i.e. minimizes the power consumption of system.
- ✓ Unit cost and cost NRE (Non Recurring Engineering)

Characteristics to Embedded System



Code size: The software or firmware code size should be optimized to accommodate maximum features with minimum memory requirement.

Efficiency: Embedded systems should use memory and power efficiently to provide high performance and long life.

High performance: Performance should be high enough to complete the specific tasks within stipulated time.

User Interface, Reliability & Cost effective.

Designed for long life with lower maintenance cost.



Common Design Metrics

NRE cost (Nonrecurring Engineering cost): it represents the monetary cost that will not be happening again and again mainly the design cost

Performance: the execution time of the system or throughput of the system

Power: the amount of power consumed by the system , which determines the life time of the battery

Flexibility: the ability to change the functionality of the system without incurring heavy NRE cost

Unit cost: represents the monetary cost spent on manufacturing each copy excluding NRE cost

Size: The physical space required by the system, measured in bytes for software, and gates or transistors for hardware



Common Design Metrics

Time-to-prototype: the time required to build a working model of the system, to refine the system's functionality

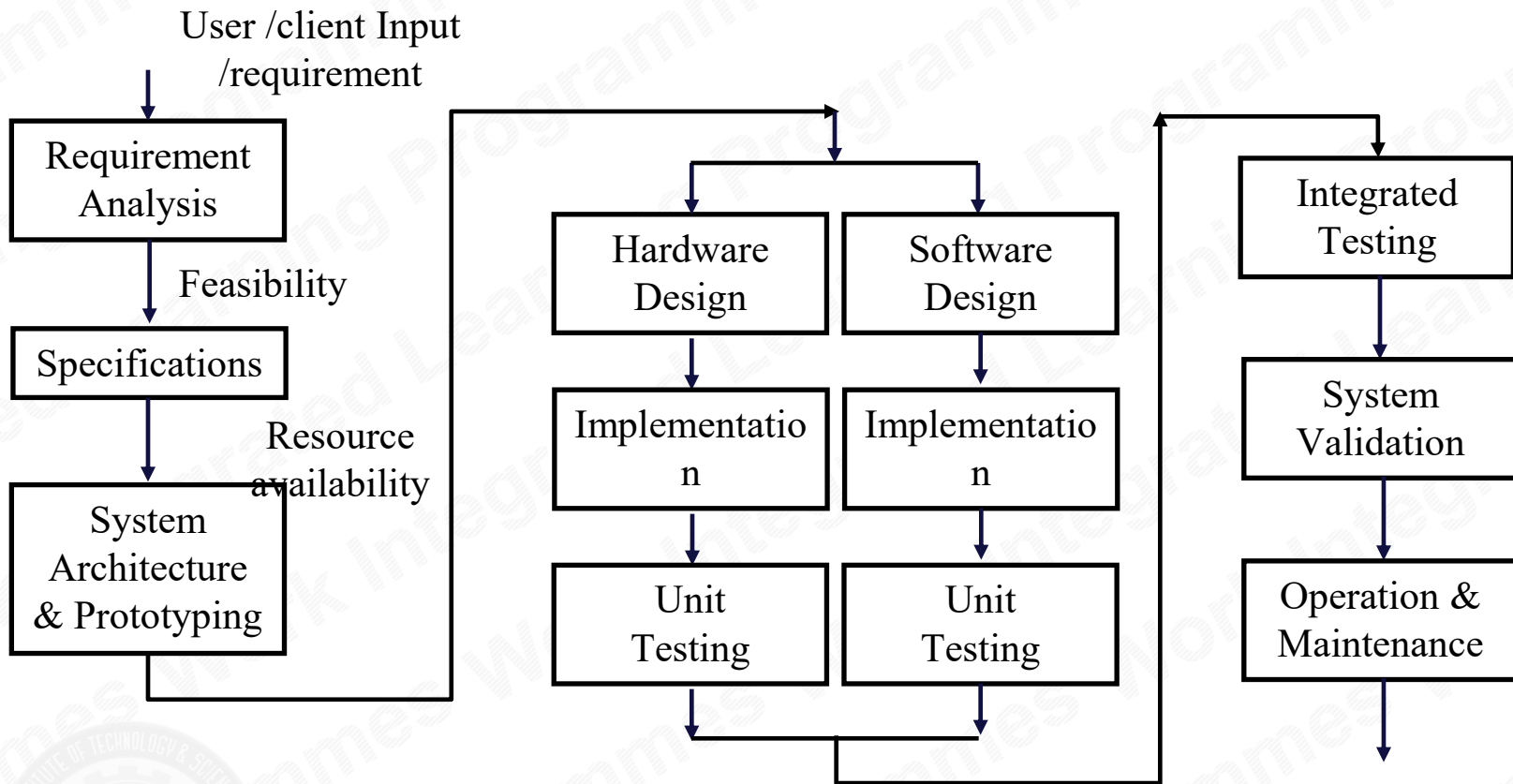
Time-to-Market: The time required to develop the system to the point that it can be released and sold to customers. The main contributors are design time, manufacturing time and testing time.

Maintainability: the ability to modify the product design after manufacturing. especially by the designers who did not do it originally

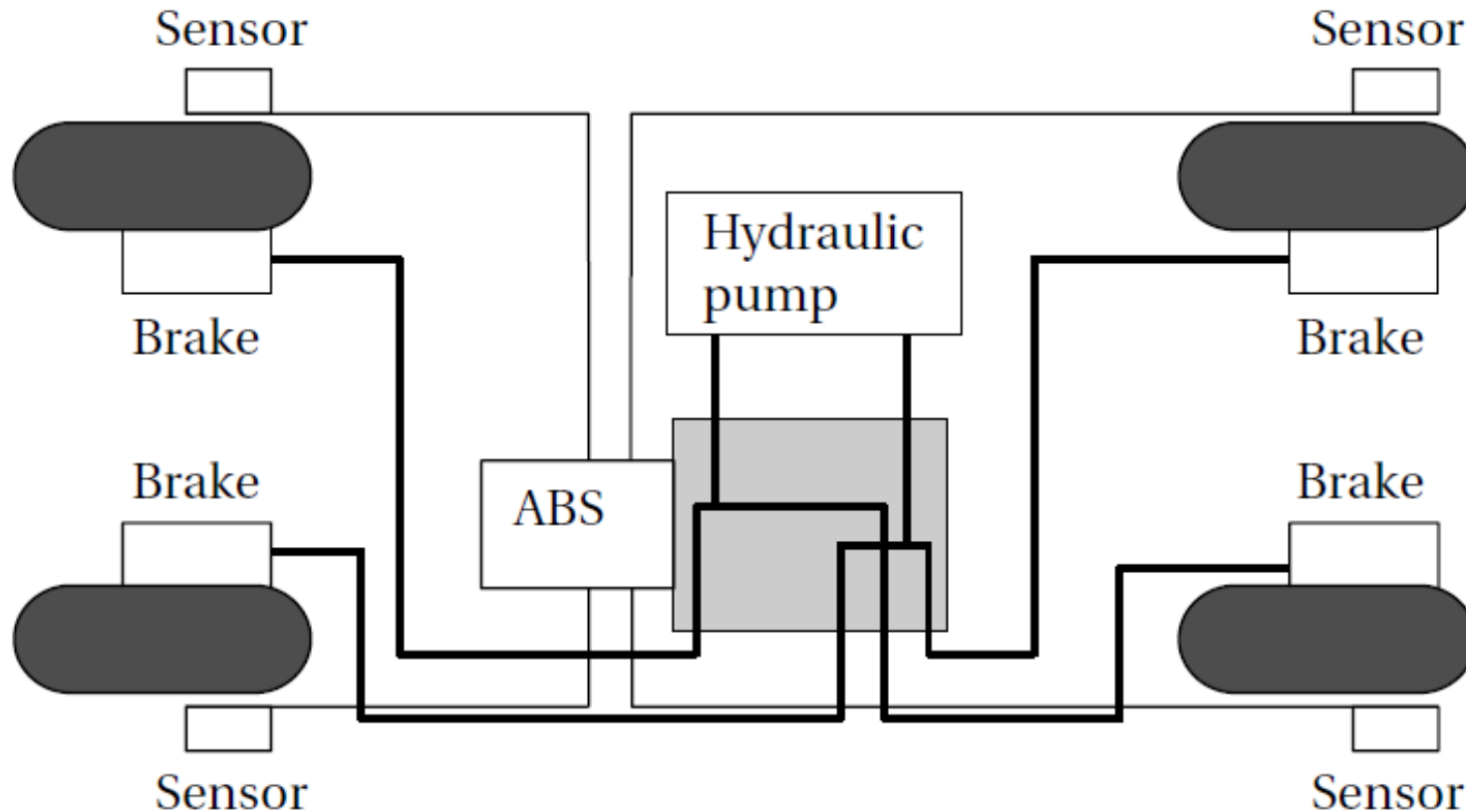
Correctness: Testing and verification

Safety: the probability that the system will not cause harm

Introduction to Embedded System



Example of ES - ABS System



END
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Session – 1
Thank You

